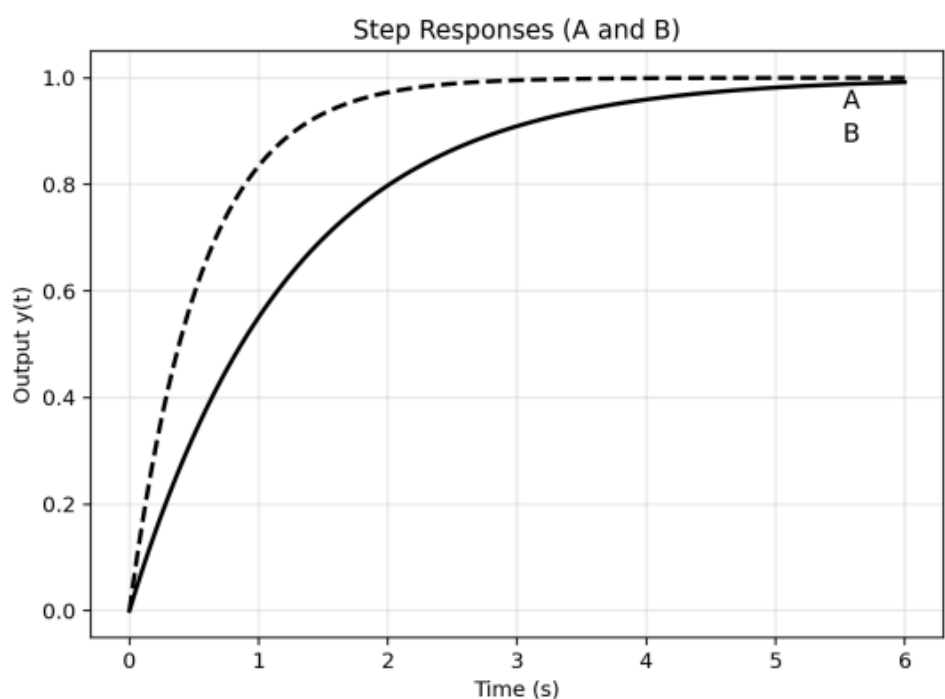
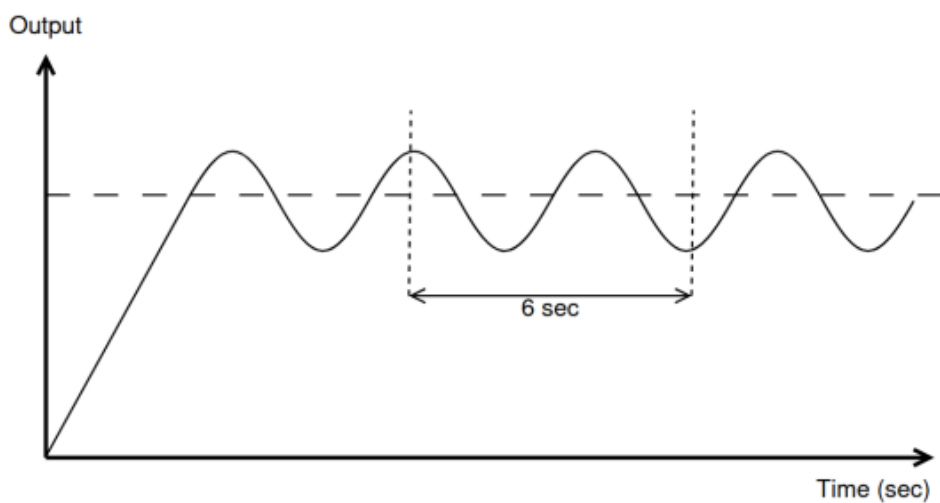


Question 1: See the diagram below and write the relations (e.g., >, <, =, faster/slower) in the blanks.



Rise time (T_r): A < B
Settling time (T_s): A < B

Question 2: Consider the following sustained oscillation figure.



Question 2 (continued):

For this figure the ultimate gain was found as **K**. After finding the ultimate gain K, your CSE461 course instructor uses only P controller. When you curious mind ask the instructor what is the value of K_p using the Ziegler–Nichols method, he gave you a hint in the below

Hint: $K_p = 0.611K - 1$

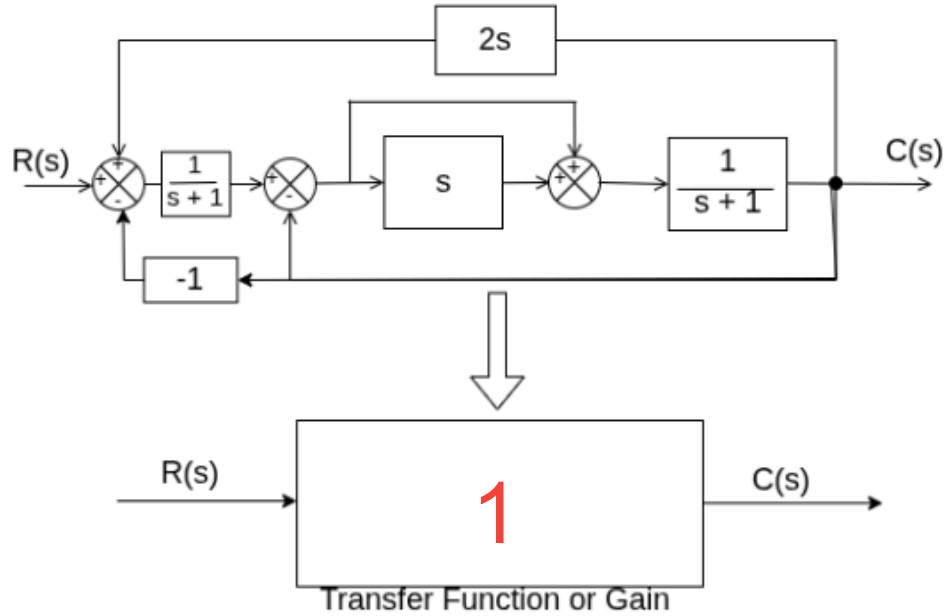
Write the values:

Ultimate Gain, K = 9
Ultimate Period, P = 4 s

Now you want to use a PID controller for the same system. Therefore what will be your K_p , K_i and K_d values using Ziegler–Nichols method?

K_p = 5.4
 K_i = 0.5
 K_d = 0.5

Question 3: Block diagram reduction.



(a) Write down the Transfer Function or Gain of the whole system in the given box.

(b) If the Transfer Function or Gain = ∞ , what will be the value of s?

s = None

Question 4: PID control for an AC system.

An air conditioning (AC) system is currently set to 18°C. You feel too cold, so you want to increase the temperature to 25°C using a PID controller. Given $K_p = 0.1$, $K_i = 0.02$, $K_d = 0.01$. The error at step 0 is $e_0 = 0$. Please write down the next 5 steps in the table below.

Step	Error	PID value	Current Temperature
1	7	0.91	18.91°C
2	6.09	0.8617	19.77°C
3	5.23	0.8808	20.65°C
4	4.35	0.8796	21.53°C
5	3.47	0.8610	22.39°C